

Supporting Information

Dual-Function Catalytic Regulation of Polysulfides by V_2CT_x/SnS_2 Heterostructures Enabling High-Performance Li–S Battery Separators

Longyu Liu^{a,b,†}, Fei Xing^{a,b,*,†}, Songshi Xu^b, Yingfei Wen^b, Minghui Dong^c, Qi Fan^c, Ze Zhang^{a,d,e,f,*}

^aCollege of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China.

^bSchool of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China.

^cSchool of Materials Science and Engineering, Southeast University, Nanjing, China.

^dAerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

^eKey Laboratory of Computational Optical Imaging Technology, Chinese Academy of Sciences, Beijing, China.

^fQilu Aerospace Information Research Institute, Jinan, China.

*E-mail: Fei Xing(xingfei@sdut.edu.cn) and Ze Zhang(zhangze@aircas.ac.cn)

[†]Longyu Liu and Fei Xing contributed equally to this work.

S1. Experimental and Computational Details**S1.1 Scanning Electron Microscopy (SEM)**

SEM images were acquired using a Hitachi SU8010 cold-field-emission scanning electron microscope. Samples were mounted on conductive adhesive on a specimen holder and sputter-coated with gold before imaging.

S1.2 Transmission Electron Microscopy (TEM)

High-resolution TEM images were acquired using a Philips Tecnai F20 instrument. Samples were dispersed in anhydrous ethanol by ultrasonication, deposited onto an ultrathin support grid, and dried before characterization.

S1.3 X-Ray Diffraction (XRD)

XRD patterns were collected using a Rigaku MiniFlex 600 powder diffractometer with Cu K α radiation ($\lambda = 1.54060 \text{ \AA}$). Data were recorded over a 2θ range of $5\text{--}80^\circ$ at a scan rate of $10^\circ \text{ min}^{-1}$.

S1.4 Ultraviolet–Visible (UV–Vis) Absorption Spectroscopy

A Thermo Fisher UV–Vis spectrophotometer was used to evaluate the residual polysulfide concentration after adsorption by the different materials. Spectra were recorded over 300–550 nm.

S1.5 X-Ray Photoelectron Spectroscopy (XPS)

XPS measurements were performed using a Thermo Scientific K-Alpha instrument with Al K α radiation ($h\nu = 1486.6 \text{ eV}$) and a $400 \text{ }\mu\text{m}$ beam. Binding energies were calibrated to the C 1s peak at 284.8 eV , and peak fitting was performed using Avantage software.

S1.6 First-Principles Calculations

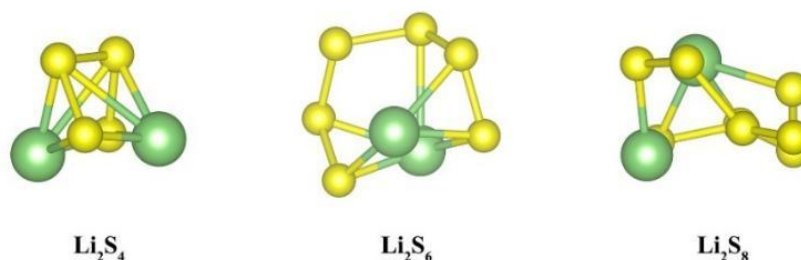
All calculations were performed within density functional theory using the projector augmented-wave method, as implemented in the Vienna Ab initio Simulation Package (VASP).^[1] The Perdew–Burke–Ernzerhof generalized-gradient approximation was used for the exchange–correlation

functional.^[2] Long-range van der Waals interactions were described using the DFT-D3 method.^[3] The plane-wave cutoff energy was 500 eV, and the electronic self-consistency criterion was 10^{-5} eV. A 15 Å vacuum layer was introduced perpendicular to the sheet to minimize interactions between periodic images. Brillouin-zone integration employed a $1 \times 1 \times 1$ k-point mesh. All structures were relaxed until the residual force on each atom was below 0.02 eV Å^{-1} .

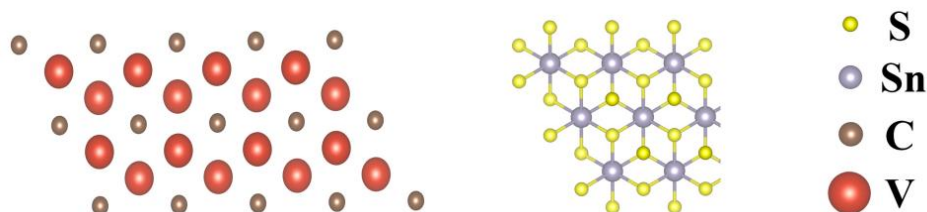
The adsorption energy (E_{ads}) was calculated using Equation S1:

$$E_{\text{ads}} = E_{\text{total}} - E_{\text{sub}} - E_{\text{adso}} \quad (\text{S1})$$

where E_{total} is the total energy of the optimized adsorbate–substrate system, and E_{sub} and E_{adso} are the energies of the isolated substrate and adsorbate, respectively, calculated in the same supercell.



Scheme S1. Computational models of the ground-state Li_2S_4 , Li_2S_6 , and Li_2S_8 clusters.



Scheme S2. Top views of the computational models of (a) V_2C and (b) SnS_2 .

S1.7 Coin-Cell Assembly

CR2032 coin cells were assembled in an argon-filled glovebox with H_2O and O_2 concentrations below 0.1 ppm. Lithium-metal foil, a KB@S cathode, and a pristine or modified polypropylene separator (Celgard 2500) were used. The electrolyte consisted of 1.0 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and 2 wt% anhydrous LiNO_3 dissolved in 1,3-dioxolane/1,2-dimethoxyethane (DOL/DME; 1:1, v/v). A total of 40 μL of electrolyte was used per cell (20 μL on each side of the separator). The sulfur loading was approximately 1.1 mg cm^{-2} . The assembly sequence was negative case, lithium-metal foil, electrolyte, separator, electrolyte, cathode, spacer, spring, and positive case, followed by pressure sealing. Cells were rested before testing, and electrochemical measurements were performed at $25 \pm 2^\circ\text{C}$.

S1.8 Cyclic Voltammetry (CV)

CV measurements of Li–S cells and symmetric cells were performed using an IVIUM electrochemical workstation. Li–S cells were scanned from 1.7 to 2.8 V at $0.1\text{--}0.5 \text{ mV s}^{-1}$. Symmetric cells were scanned from -1.0 to 1.0 V at 10 mV s^{-1} .

S1.9 Electrochemical Impedance Spectroscopy (EIS)

EIS measurements were conducted using an IVIUM Vertex.C electrochemical workstation over a frequency range of 100 kHz–0.1 Hz with a perturbation amplitude of 5 mV.

S1.10 Galvanostatic Charge–Discharge Testing

Galvanostatic charge–discharge measurements were performed using a Neware CT-4008Tn battery-testing system over 1.7–2.8 V. A specific current of 1 C = 1675 mA g^{−1} was used for rate conversion. The reported metrics include specific capacity, Coulombic efficiency and capacity-decay rate.

S2. Supporting Figures

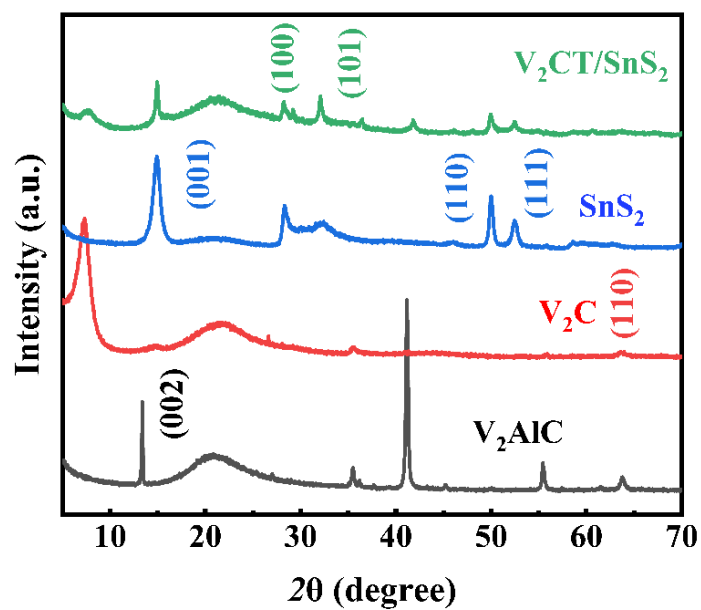


Figure S1. XRD patterns of V_2AlC , V_2C , SnS_2 and the V_2CT_x/SnS_2 composite.

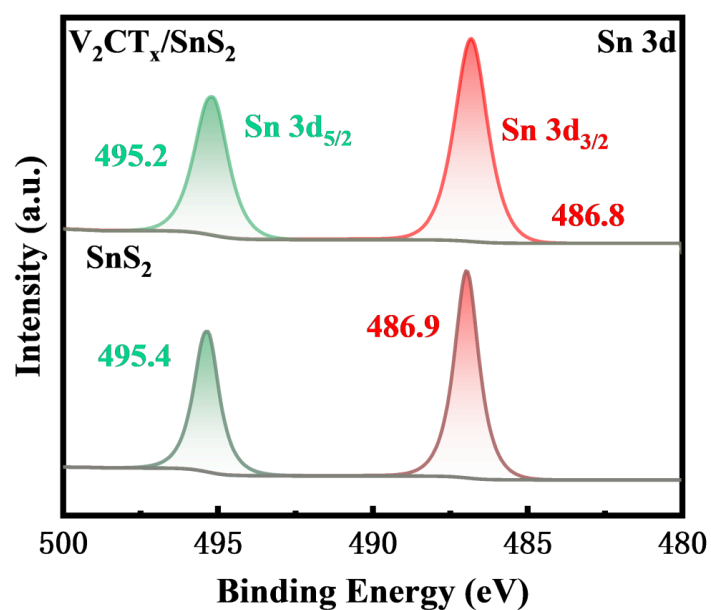


Figure S2. High-resolution Sn 3d XPS spectra of SnS_2 and V_2CT_x/SnS_2 .

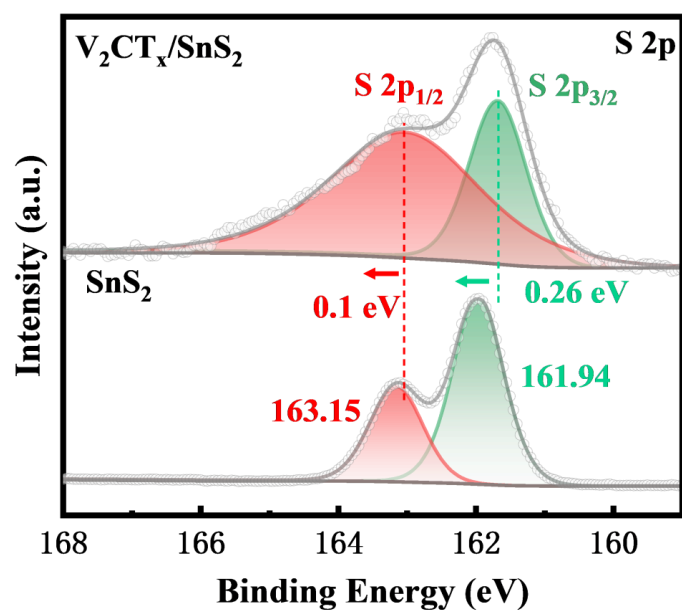


Figure S3. High-resolution S 2p XPS spectra of SnS_2 and $\text{V}_2\text{CT}_x/\text{SnS}_2$.

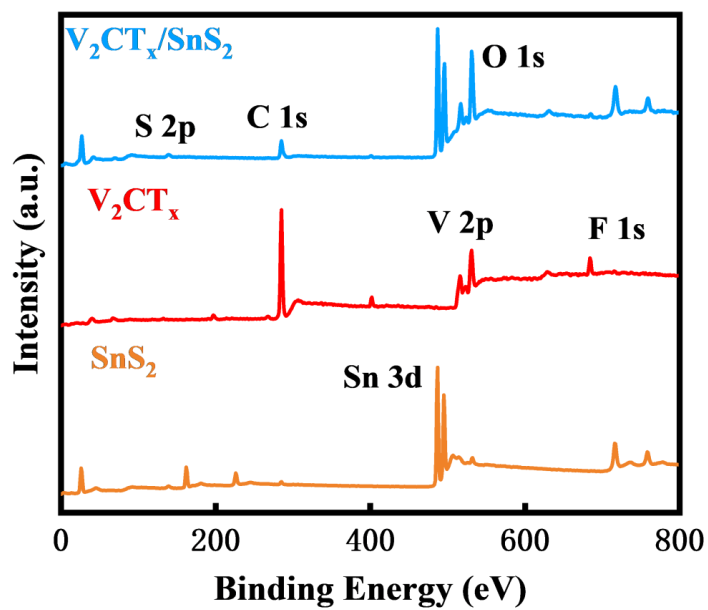


Figure S4. XPS survey spectra of SnS_2 , V_2CT_x and $\text{V}_2\text{CT}_x/\text{SnS}_2$.

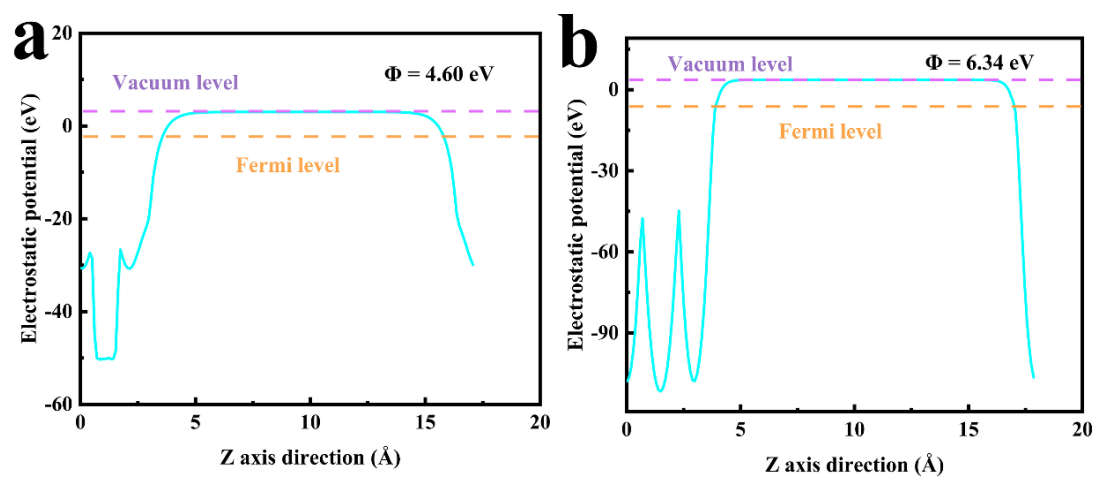


Figure S5. Electrostatic-potential distributions of (a) V_2CT_x and (b) SnS_2 .

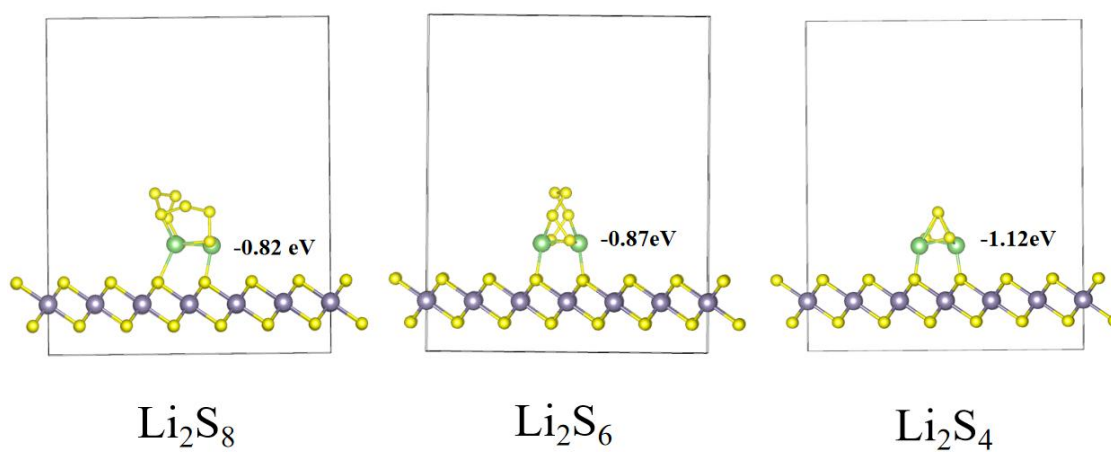


Figure S6. Calculated adsorption energies of Li_2S_8 , Li_2S_6 and Li_2S_4 on SnS_2 .

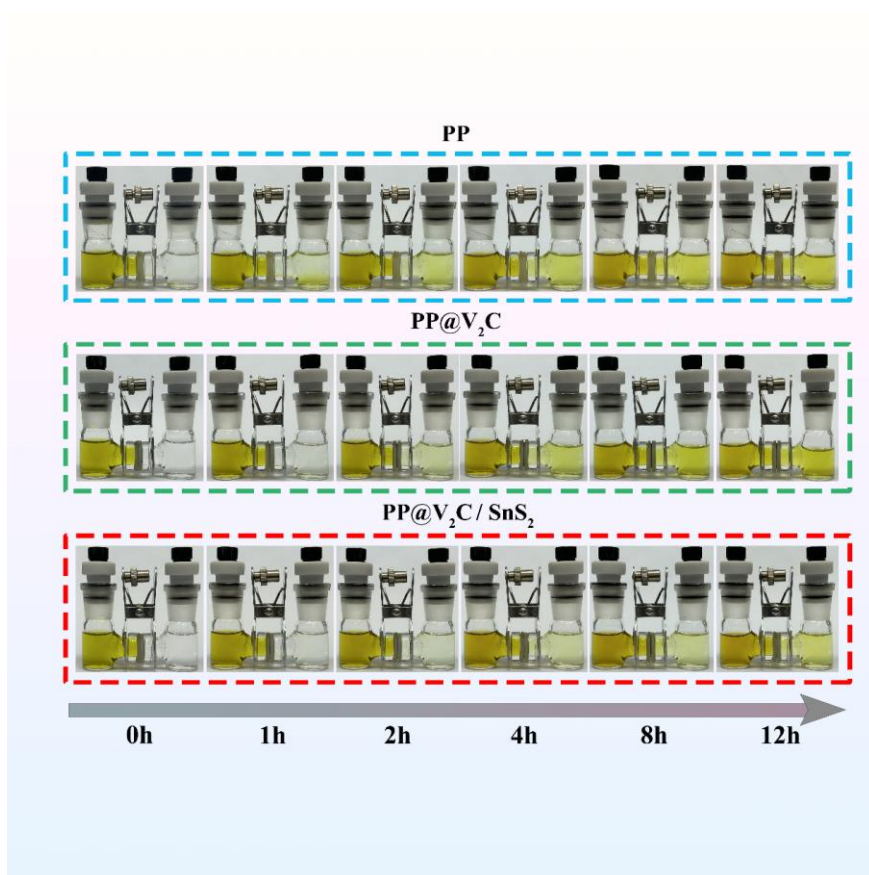


Figure S7. Photographs of Li_2S_6 diffusion through PP, $\text{PP@V}_2\text{CT}_x$ and $\text{PP@V}_2\text{CT}_x/\text{SnS}_2$ separators at different times in an H-type cell.

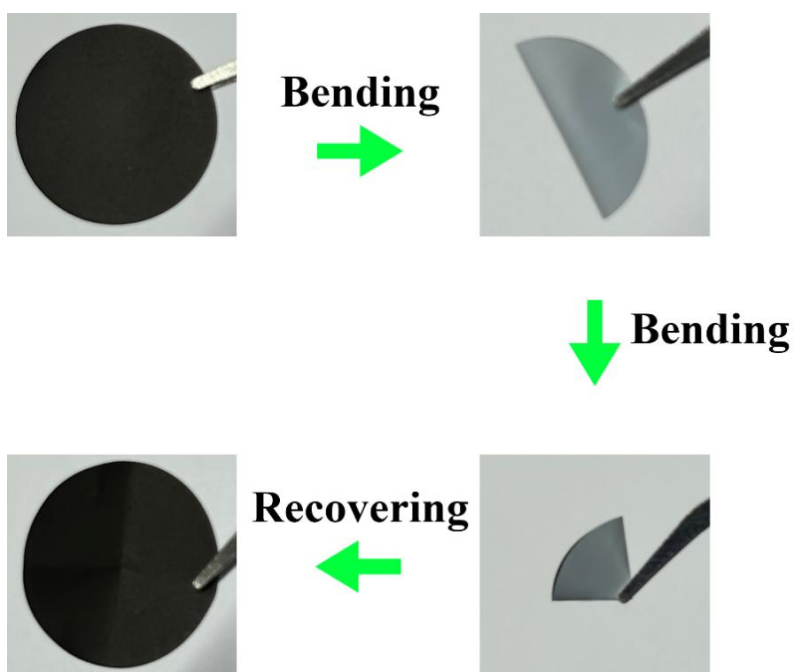


Figure S8. Photographs of the $\text{PP@V}_2\text{CT}_x/\text{SnS}_2$ separator in the bent state and after recovery.

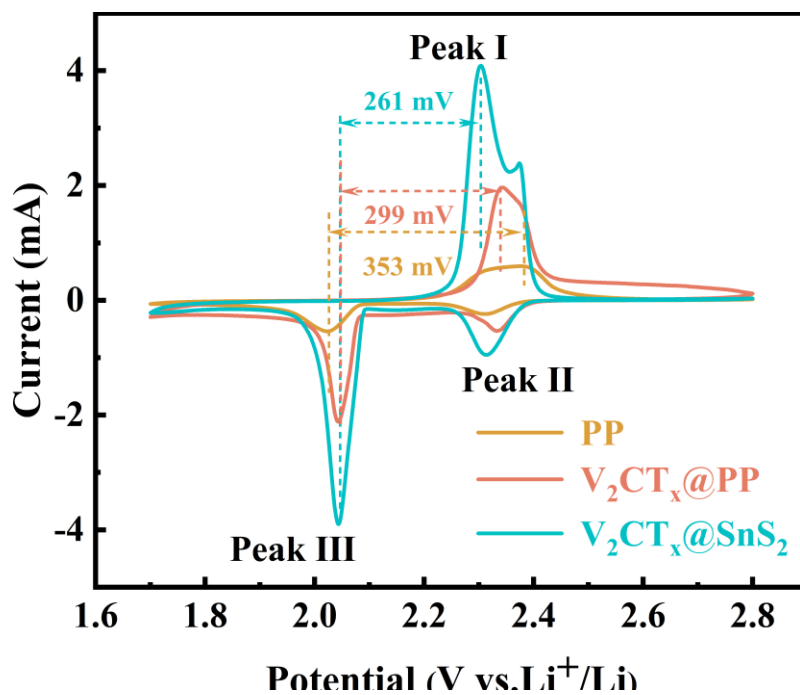


Figure S9. CV curves of cells with different separators at 0.1 mV s^{-1} .

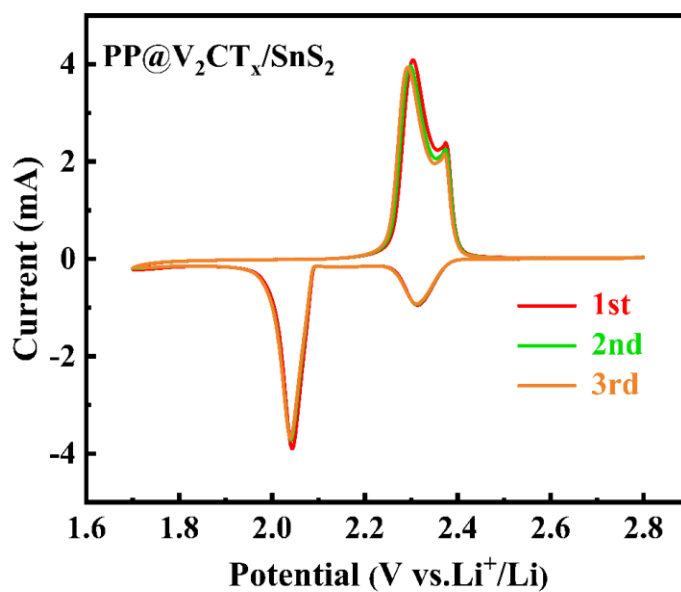


Figure S10. CV curves for the first three cycles of the $PP@V_2CT_x/SnS_2$ cell at 0.1 mV s^{-1} .

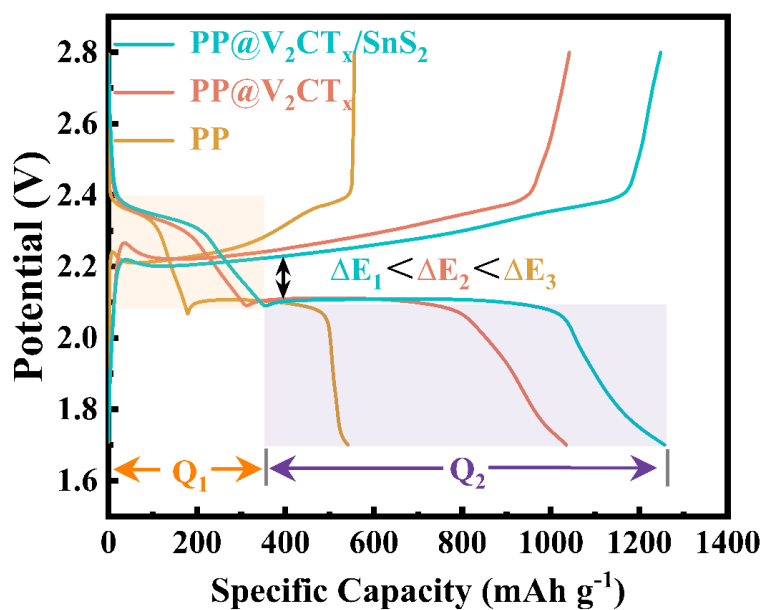


Figure S11. Galvanostatic charge–discharge profiles at 0.2 C.

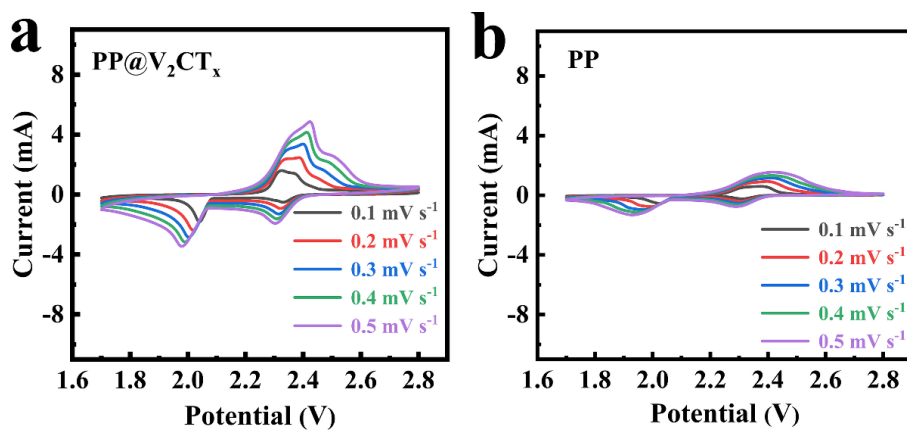


Figure S12. CV curves recorded from 0.1 to 0.5 mV s^{-1} for cells with (a) $\text{PP@V}_2\text{CT}_x$ and (b) PP separators.

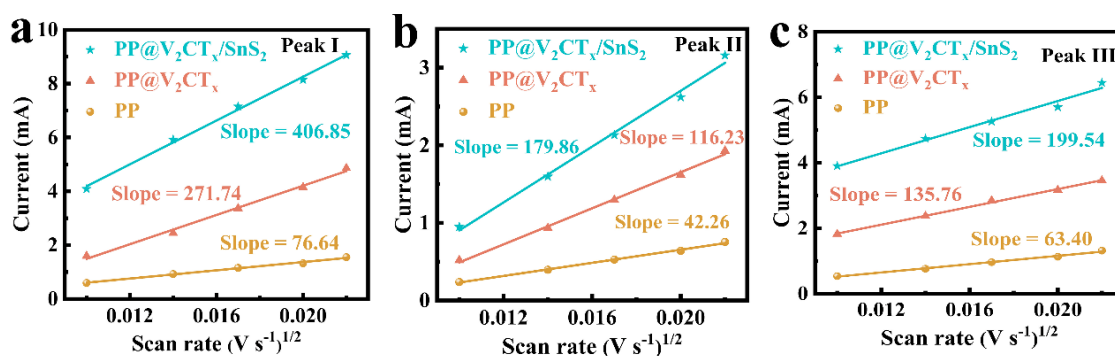


Figure S13. Linear fits of redox peak current versus the square root of scan rate for Peaks I–III, derived from Figure 3b and Figure S12a,b.

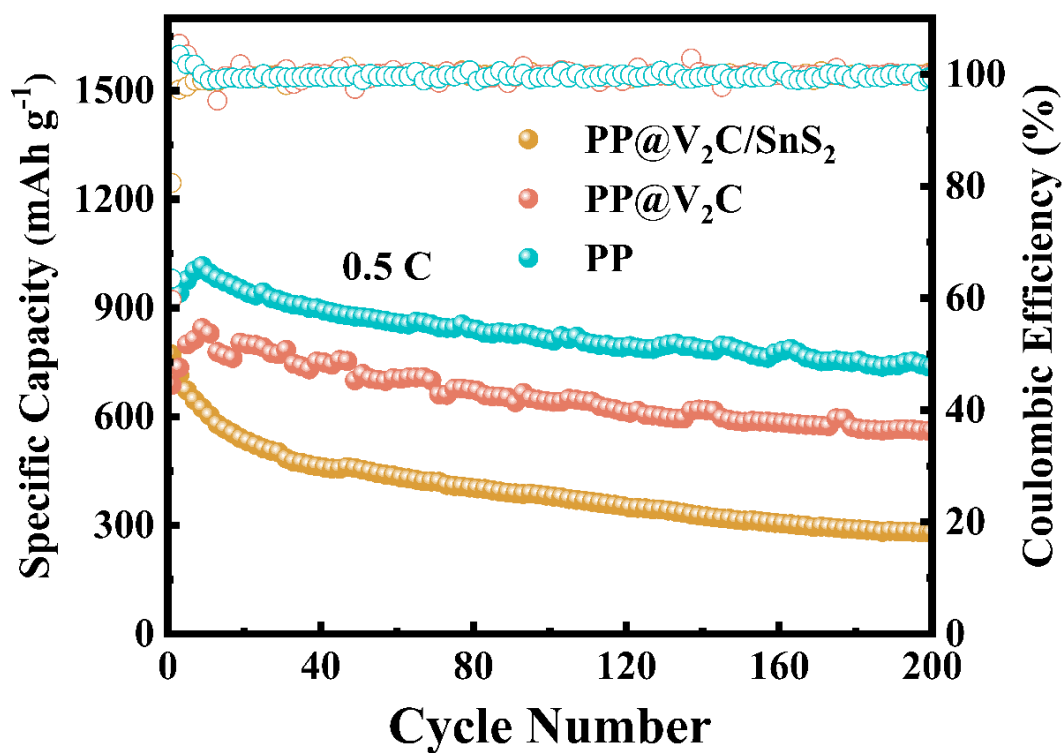


Figure S14. Cycling performance of Li–S cells with different separators at 0.5 C.

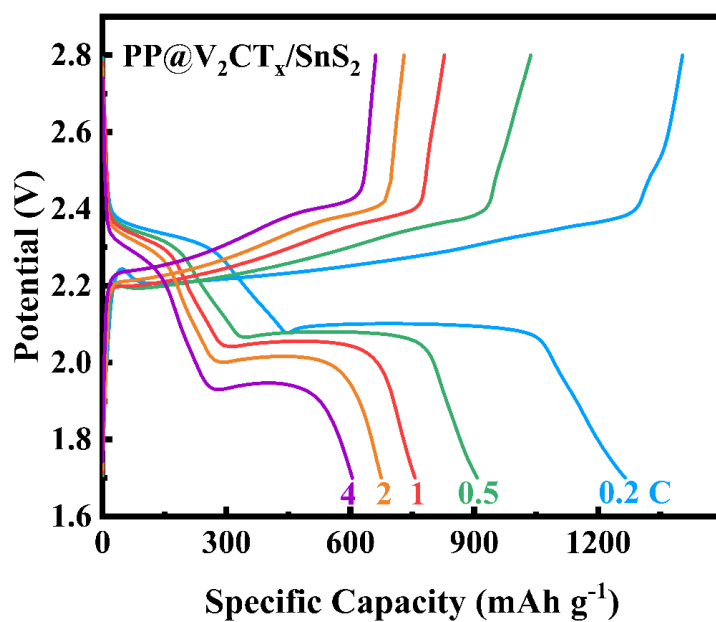


Figure S15. Galvanostatic charge–discharge profiles of the V_2CT_x/SnS_2 -separator cell at different rates.

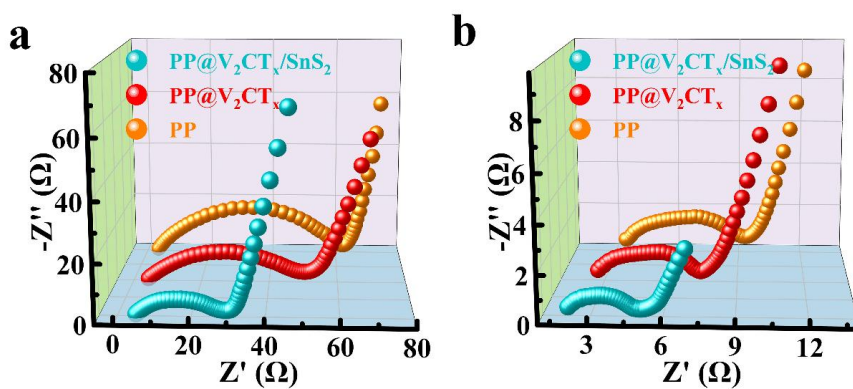


Figure S16. Nyquist plots of cells with different separators (a) before cycling and (b) after 1000 cycles at 1 C.

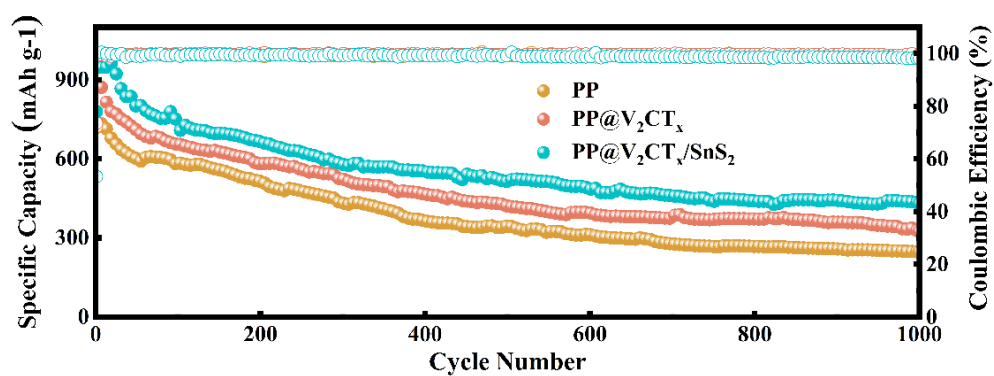


Figure S17. Long-term cycling performance of Li-S cells with different separators at 1 C.

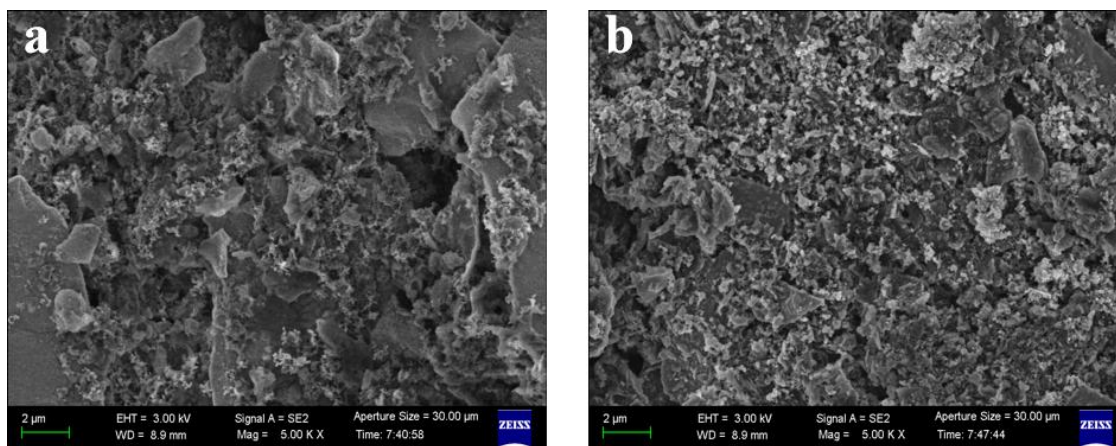


Figure S18. Top-view SEM images of the V₂CT_x/SnS₂/PP membrane (a) before and (b) after cycling.

S3. Supporting Table

Table S1. Comparison of the electrochemical performance of the present separator with previously reported systems.

Separator material	Sulfur loading (mg cm ⁻²)	Rate / capacity (C / mAh g ⁻¹)	Decay rate (% per cycle)
Ti ₃ C ₂ T _x /CNTs (10%) ^[4]	1.2	2 C/640	0.086
TOS/MX/TOS ^[5]	–	1 C/804.5	0.048
FeSA-PN@PNC ^[6]	1.5	2 C/659	0.040
MoS ₂ -x-Co ₉ S ₈ -y/rGO ^[7]	0.8–0.9	3 C/710.2	0.060
N-MX-CoS ₂ ^[8]	1.2	4 C/775	0.052
Ti ₃ C ₂ T _x /Nafion ^[9]	2	3 C/794	0.03
PM(0.4M)-CNT ^[10]	0.91	2 C/677.6	0.07
Ta ₄ C ₃ -Ta ₂ O ₅ @C ^[11]	1.2	1 C/738.5	0.086
OMC-g-MXene ^[12]	1.2	5 C/537	0.047
Co-MoS ₂ @NC ^[13]	1	4 C/610.1	0.053
STO-W/S ^[14]	0.8–1	2 C/492	0.067
MPF13-550/PP ^[15]	1.2–1.6	2 C/593	0.375
CoSe ₂ @TiSe ₂ -C ^[16]	2.5	3 C/717	0.031
7ZnO-3ZnS/rGO ^[17]	1	3 C/748.9	0.06
RG@CoS@C ^[18]	3	2 C/677	0.08
TiO ₂ @MoS ₂ ^[19]	1.0–1.2	5 C/601	0.067
VC-1 ^[20]	2.4–3	2 C/733.3	0.041
CC@VS ₂ -VO ₂ @Li ₂ S@C ^[21]	1.24	3 C/462.8	0.120
This work	1.08	4 C/653.0	0.019

S4. References

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